

IS THE HOME–FIELD ADVANTAGE IN LIMITED OVERS ONE–DAY INTERNATIONAL CRICKET ONLY FOR DAY MATCHES?

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Key words: Winning the toss; home field advantage; cricket; logistic regression; quantile regression.

Summary: The impact of the factors “home field advantage”, “winning the toss” and “team superiority” on the outcome of Limited Overs One Day International (ODI) cricket matches is investigated. Our results show that the “home-field-advantage” is a significant factor but only for “Day” matches. Confirming previous studies, we also show that “winning the toss” does not give any statistically significant advantage towards the outcome of the game.

1. Introduction

Cricket is a field game played by two competing teams. The game is played on a grass field in the center of which is a flat strip of ground measuring 20 meters long and 3 meters wide, which is called the pitch. At both ends of the pitch, 20 meters apart, “wickets” are placed. A wicket, usually made of wood, is used as the target for a bowler who throws a stiff-arm pitch. A batsman tries to hit the ball with a cricket bat while also protecting his wicket. A team’s objective is to dominate its opponent by scoring more runs. Readers can find much more detailed information about the game of cricket and its regulations on the web at <http://www.espnccricinfo.com>, but there are many other common sources of information about cricket.

There are mainly three types of cricket matches played today: Test cricket, One Day International (ODI) and Twenty20 International (T20I), which is rapidly becoming popular among cricket fans. Test cricket matches usually last up to five days and each team is allowed to play up to two innings during these five days. One Day International (ODI) matches, also the focus in our study, are played on a single day. In this format each team gets only one innings, which is restricted to a maximum number, usually fifty overs (50×6 legal ball deliveries). Each innings ends after 50 overs, no matter how many batsmen are out. If ten batsmen are out before the 50th over, the innings is also over. If the innings of the team which bats first ends in this manner, the second team still has its full number of overs to score runs. In each innings, each bowler is restricted to bowling a maximum number of overs equal to one-fifth of the total number of overs in the innings. There are two types of ODI cricket matches: “Day” matches, which are played and completed during the daytime, and “Day & Night” matches where one team bats during the afternoon while the other bats in the evening.

In Twenty20 cricket, each team gets a single innings to bat for a maximum of 20 overs. A match is completed in approximately three hours, with innings lasting about 75 minutes, bringing the game time closer to that of other popular team sports, like soccer and baseball. In all of these cricket formats, each team consists of eleven players; however, there are other formats, like six players per team, which are uncommon in international cricket.

Several studies in the literature have shown the significance of “home field advantage” with respect to the outcome of limited-overs cricket matches. Some of these studies have also shown that “winning the toss” does not give any significant advantage for ultimate victory in these matches. In this article we discuss some surprising new findings related to the importance of “home-field-advantage” and “winning the toss” by analyzing limited-overs cricket data separately for “Day” and “Day & Night” matches. Specifically, the analyses presented here use the tools of Logistic regression and Quantile regression applied to data obtained from previous matches.

Like other sports, such as soccer (Clarke and Norman, 1995), baseball (Levernier and Barilla, 2007), volleyball (Marcelino, Mesquita, Palao and Sampaio, 2009), and netball (Pledger and Morton, 2010), the existence of “home-field-advantage” is a common belief among the cricket-playing community. Various explanations can be found in the literature for possible reasons for “home field advantage,” such as familiarity with the home field, local weather conditions, the effect of the predisposition of fans, and reduced travel stress, all of which we collectively refer to here as environmental conditions. Allsopp and Clarke (2004) and De Silva and Swartz (1997) showed that teams that play at home have a significant advantage over their opponents. De Silva and Swartz (1997) also estimated the marginal increase in winning percentage for matches played at home. De Silva and Swartz (1997) have noted that some teams seem to have the “home-field-advantage” at so-called “neutral” venues. For example, when Australia and Sri Lanka play in India, Sri Lanka appears to benefit from the “home-field-advantage” due to the popularity of the Sri Lankan team among the predominantly Indian fans as well as the adaptability of the Sri Lanka players to familiar environmental conditions.

With respect to “winning the toss”, the team captain who wins the toss can elect for his team to either “bat” or “field” first. Such a choice is typically based on knowledge of the team’s various strengths and weaknesses, and other factors such as environmental conditions. De Silva and Swartz (1997) and Allsopp and Clarke (2004) have shown that, contrary to common belief, there is no significant effect on the outcome of a match by “winning the toss”. Even though there are some

matches where “winning the toss” seemed to have a large role in game outcome, there is little statistical evidence (De Silva and Swartz, 1997) to support this claim. For example, the 2007 World Cup Final featured Sri Lanka and Australia, and the media and some fans predicted that, based on the strengths of the two teams, whichever team won the toss would ultimately win the game. As it happened, Australia won the toss and subsequently won the title as well. Even though some may ignore this as a purely random event, those who predicted the outcome based on the “winning the toss” factor were pretty certain of the final outcome, even before this match was played.

2. Logistic Regression Approach

In order to better understand the factors affecting the outcome of a cricket match, we chose to use a logistic regression model to measure the significance of the possible factors. Our analysis is focused on One Day International (ODI) matches played from January 2003 through March 2011. Using ICC rankings, we studied the matches played by the top eight international teams: Australia, England, India, Pakistan, New Zealand, South Africa, Sri Lanka and West Indies. Few matches ended in a tie and those were deleted from the analysis so as to keep dichotomy in the response variable. Logistic regression was used to model the outcome of a game (win/lose) using the explanatory variables *Toss* (“winning the toss”), *Home* (“home-field-advantage”) and the *Superiority* (difference between the ICC rankings of the team and its opponent).

On letting p_i denote the probability that the team that bats first wins the match, the logistic regression model becomes

$$\ln \left(\frac{p_i}{1 - p_i} \right) = \beta_0 + \beta_1 Toss_i + \beta_2 Home_i + \beta_3 Superiority_i$$

where $o_i = \ln \left(\frac{p_i}{1 - p_i} \right)$ denotes the log-odds and the i index ranges over the sequential playing of the matches described in the previous paragraph. The analysis was conducted using SAS[®] 9.2.

2.1. Logistic Regression Results for the Top Eight ODI Teams

Tables 1 and 2 present a descriptive analysis of the ODI matches played by the top eight teams. For example, the first line of Table 1 shows that India played 205 ODI matches during the time period considered: 58.5% were “Day & Night” matches and 41.5% were “Day” matches. In total, 39% were played at home, 40% were played at the opponent’s home, and 21% were played at neutral venues, and accompanying winning percentages were 62.5% (home), 45.1% (opponent’s home), and 51.2% (neutral). The first line in Table 2 shows that India won the toss in 51.2% of the matches and then went on to win 58.1% of these matches. But, India won 48.0% of the matches after having lost the toss. Also, India chose to bat first in 61.0% of matches when it won the toss. On the other hand, India was sent in to bat first in 34.0% of matches when it lost the toss. Regarding “Day” matches, India batted first in 44.4% of these matches when it won the toss, but was sent in to bat in 52.5% of these matches when it lost the toss. On the other hand, for “Day & Night” matches, India batted first in 73.3% of these matches when it won the toss, but was sent in to bat in 21.7% of these matches when it lost the toss. Subsequent lines in these tables are read similarly.

Table 1: Descriptive Percentages for Matches Played by Venue.

Country	Matches Played			Venue			Matches Won		
	Total	Day %	Day & Night %	Home %	Opponent's Home %	Neutral %	Home %	Opponent's Home %	Neutral %
India	205	41.5	58.5	39.0	40.0	21.0	62.5	45.1	51.2
New Zealand	150	44.7	55.3	41.3	30.0	28.7	58.1	28.9	48.8
Sri Lanka	161	41.0	59.0	37.9	37.9	24.2	59.0	41.0	46.2
England	153	52.3	47.7	45.1	41.8	13.1	47.8	32.8	40.0
Australia	201	34.8	65.2	40.8	40.3	18.9	72.0	61.7	86.8
Pakistan	158	34.8	65.2	27.2	41.1	31.6	51.2	41.5	48.0
South Africa	136	44.8	55.2	38.2	41.9	19.9	69.2	49.1	44.4
West Indies	130	62.3	37.7	40.8	36.2	23.1	30.2	27.7	23.3

Table 2: Toss and Batting Percentages for all Eight Teams.

Country	Toss						Batted First			
	Toss Won			Toss Lost			Day		Day & Night	
	%	Batted First %	Match Won %	%	Batted First %	Match Won %	Toss Won %	Toss Lost %	Toss Won %	Toss Lost %
India	51.2	61.0	58.1	48.8	34.0	48.0	44.4	52.5	73.3	21.7
New Zealand	53.3	43.8	52.5	46.7	47.1	40.0	22.2	67.7	61.4	30.8
Sri Lanka	47.2	73.7	52.6	52.8	44.7	45.9	51.9	71.8	85.7	21.7
England	51.0	55.1	43.6	49.0	42.7	37.3	30.8	56.1	79.5	26.5
Australia	47.2	83.3	66.7	52.8	41.0	74.3	71.9	60.5	89.1	29.8
Pakistan	51.3	61.7	44.4	48.7	32.5	48.0	40.6	73.9	75.5	14.8
South Africa	51.5	57.1	55.7	48.5	31.8	56.1	31.0	53.1	75.6	11.8
West Indies	46.9	36.1	32.8	53.1	44.9	23.2	20.9	63.2	72.2	22.6

Teams generally decide to bat first when they win the toss mainly because of their stronger batting ability or because of the unpredictability of the second half of the game due to the environmental conditions. This unpredictability is the situation for “Day & Night” matches. Descriptive details of the matches played by each of these teams can be explained similarly.

Table 3 shows SAS[®] logistic regression output for the Indian team analysis using the entire data set where both “Day” and “Day & Night” matches are included together. As seen, *Superiority* and *Home* are highly statistically significant. However, *Toss* is not significant, which is contrary to the opinion of cricket fans, but which is, nonetheless, consistent with the literature.

Table 4 shows results for “Day & Night” matches only. In this case of India, *Toss* is not significant but *Superiority* is, again, statistically significant. In contrast to the previous case *Home* is not a significant factor. Additionally, Table 5 shows logistic regression results for “Day” only matches. *Superiority* and *Home* are statistically significant factors for India, but *Toss* is not. Again, this is consistent with the literature but contrary to fan belief.

The boldface *P*-values in Tables 3 – 5 are associated with the statistically significant factors. In almost all cases *Superiority* is a statistically significant effect. The results for India, Sri Lanka and New Zealand merit further consideration. For each, *Toss* is not significant and *Home* is significant only for “Day” matches, but not for “Day & Night” matches. *Superiority* is a significant factor for both match formats. In the case of Sri Lanka, *Home* is not a significant factor when we consider both “Day” and “Day & Night” combined together. This could possibly be explained by the fact that a majority of the matches are currently “Day & Night”, whereas the majority of matches use to

Table 3: Logistic Regression Results for COMBINED ODI Matches.

Country	COMBINED DAY and DAY & NIGHT					
	Toss		Home		Superiority	
	Odds Ratio	P-value	Odds Ratio	P-value	Odds Ratio	P-value
India	1.436 (0.794,2.598)	0.2311	1.499 (1.070,2.100)	0.0186	1.053 (1.028,1.078)	<0.0001
New Zealand	1.275 (0.614,2.649)	0.5151	1.617 (1.041,2.511)	0.0325	1.072 (1.040,1.105)	<0.0001
Sri Lanka	1.570 (0.804,3.065)	0.1863	1.275 (0.869,1.872)	0.2145	1.049 (1.022,1.077)	0.0004
England	1.836 (0.899,3.749)	0.0955	1.442 (0.991,2.099)	0.0556	1.057 (1.024,1.090)	0.0005
Australia	0.756 (0.394,1.450)	0.3993	1.172 (0.817,1.682)	0.3878	1.081 (1.043,1.120)	<0.0001
Pakistan	0.896 (0.460,1.749)	0.7485	1.255 (0.835,1.887)	0.2749	1.058 (1.029,1.088)	<0.0001
South Africa	0.870 (0.418,1.812)	0.7094	1.535 (1.014,2.323)	0.0428	1.047 (1.012,1.083)	0.0084
West Indies	1.404 (0.612,3.222)	0.4233	1.072 (0.668,1.719)	0.7737	1.047 (1.008,1.086)	0.0168

Table 4: Logistic Regression Results for Day & Night Matches.

Country	DAY & NIGHT					
	Toss		Home		Superiority	
	Odds Ratio	P-value	Odds Ratio	P-value	Odds Ratio	P-value
India	1.991 (0.914,4.338)	0.0829	1.135 (0.729,1.766)	0.5755	1.052 (1.021,1.085)	0.0011
New Zealand	1.294 (0.504,3.324)	0.5917	1.383 (0.788,2.430)	0.2589	1.055 (1.014,1.099)	0.0085
Sri Lanka	1.625 (0.665,3.972)	0.2870	0.846 (0.483,1.482)	0.5596	1.060 (1.023,1.099)	0.0013
England	2.084 (0.695,6.247)	0.1898	1.265 (0.702,2.279)	0.4337	1.067 (1.012,1.125)	0.0164
Australia	0.700 (0.312,1.573)	0.3886	0.906 (0.576,1.426)	0.6699	1.110 (1.056,1.166)	< 0.0001
Pakistan	0.959 (0.418,2.200)	0.9217	0.881 (0.519,1.495)	0.6395	1.061 (1.025,1.098)	0.0008
South Africa	0.928 (0.357,2.414)	0.8782	1.255 (0.717,2.196)	0.4265	1.029 (0.984,1.076)	0.2102
West Indies	3.119 (0.718,13.557)	0.1292	0.535 (0.119,2.398)	0.4135	1.005 (0.943,1.071)	0.8849

be “Day” matches. When “Day” matches are dominating, *Home* could become significant for the whole group, containing both types of ODI formats.

For Australia, *Superiority* is a significant factor for “Day & Night” matches but not for “Day” matches. *Home* and *Toss* are not significant factors for both cases. This is due to the fact that Australia, a four-time world-cup winner, has been the best team in the last few years and their winning percentages are consistent whether playing at home or away. For Pakistan, *Superiority* is significant for each case. *Home* is (marginally) significant for only “Day” matches but not for “Day & Night” matches. *Toss* is significant neither for “Day” nor “Day & Night” cases. For South Africa, both *Superiority* and *Home* are significant for “Day” matches but not for “Day & Night” matches, while *Toss* is not significant for either of the ODI formats. For West Indies, *Superiority* is the only

Table 5: Logistic Regression Results for Day Matches.

Country	DAY					
	Toss		Home		Superiority	
	Odds Ratio	P-value	Odds Ratio	P-value	Odds Ratio	P-value
India	0.968	0.9464	2.358	0.0036	1.063	0.0030
	(0.373, 2.514)		(1.325, 4.198)		(1.021, 1.107)	
New Zealand	1.056	0.9305	2.291	0.0313	1.094	0.0004
	(0.313, 3.565)		(1.077, 4.871)		(1.041, 1.150)	
Sri Lanka	1.132	0.8256	2.514	0.0159	1.072	0.0185
	(0.376, 3.407)		(1.188, 5.320)		(1.012, 1.137)	
England	1.668	0.2913	1.570	0.0981	1.053	0.0135
	(0.645, 4.314)		(0.920, 2.678)		(1.011, 1.097)	
Australia	1.015	0.9820	> 999.9	0.9274	1.039	0.2654
	(0.280, 3.674)		(0, + ∞)		(0.971, 1.112)	
Pakistan	1.053	0.9363	2.648	0.0647	1.157	0.0106
	(0.297, 3.740)		(0.942, 7.438)		(1.035, 1.295)	
South Africa	0.790	0.7062	2.358	0.0182	1.084	0.0087
	(0.232, 2.688)		(1.157, 4.805)		(1.021, 1.152)	
West Indies	0.908	0.8580	1.222	0.5431	1.077	0.0062
	(0.314, 2.622)		(0.640, 2.333)		(1.021, 1.136)	

factor that is significant for “Day” matches, and none of the factors are significant for “Day & Night” matches. In general, West Indies was not playing well during the time period that we considered, and this is clearly seen from its low winning percentages. To a certain extent, this is the opposite of what we see for Australia. When a team’s outcome is chiefly in one direction (either “winning” or “losing”), it is *Superiority* that is the dominating significant factor.

3. An Alternate Approach: Quantile Regression

Ordinary least squares (OLS) regression is, in fact, regression towards the mean. Basically, OLS investigates the mean of the response variable at given values of predictor variables. Sometimes, the goal may not be to investigate the mean response, or investigating the mean alone may not be suitable in a given context. Quantile regression models the relationship between specific quantiles of the response variable against a set of predictor variables. In OLS regression, a particular regression coefficient represents the change in the mean of the response variable due to a unit change in the predictor variable associated with it. However, a quantile regression coefficient represents the change in a specified quantile of the response variable due to a unit change in its corresponding predictor variable. Much research has been done in the area of Quantile Regression, and Koenker (2005), Koenker and Hallock (2001), Wiseman and Chatterjee (2010), Cade and Noon (2003), Furno (2010) and Bassett (2007) are some excellent sources found in the literature. Indeed, Quantile regression is a fairly new statistical concept which is becoming increasingly popular because of its wide applicability. For purposes of comparison, we consider quantile regression as a tool to investigate the effect of the outcome of a game due to “home-field-advantage”, “winning the toss” and “superiority” predictors.

The response variable studied here is the margin of victory in a given match. In the context of

quantile regression, the τ^{th} quantile of the margin of victory is modeled as

$$Q_y(\tau | \mathbf{X}) = \beta_0(\tau) + \beta_1(\tau) \textit{Toss} + \beta_2(\tau) \textit{Home} + \beta_3(\tau) \textit{Superiority}$$

where $\mathbf{X} = (\textit{Toss}, \textit{Home}, \textit{Superiority})$ is a (1×3) row vector of predictor variables. Calculating the margin of victory is straightforward for the case where the team that bats first wins the match, say *Team 1*. Then, the margin of victory is simply the difference between total scores for the two teams. However, if the team not batting first wins the game, say *Team 2*, the calculation is different. As soon as *Team 2*'s score exceeds that of *Team 1*, the game is ended and rules declare *Team 2* as the winner. There is no direct calculation for margin of victory in this situation, but we can make use of the Duckworth-Lewis (1998) method, which was developed to calculate revised target scores for interrupted matches. Note that De Silva, Pond and Swartz (2001) also used a similar approach to calculate the magnitude of the victory in ODI matches. Duckworth-Lewis (1998) constructed a table of remaining percentage resources based on the number of remaining overs and the number of remaining batsmen. Following that development, let R_2 and S_2 denote, respectively, the remaining percentage of resources and the total score for *Team 2*. The predicted total score, P_2 , for *Team 2* is calculated as if it was allowed to use 100% of its remaining resources and is $P_2 = \frac{S_2}{R_2} \times 100$. The difference between the total score for *Team 1* and this predicted score, P_2 , for *Team 2* is defined as the margin of victory for the match. Obviously, P_2 is a negative value since *Team 1* lost the match in this instance.

To adjust unusually large magnitude margin of victory values, we used the De Silva et al. (2001) modification for the Duckworth-Lewis percentage resources table. As with the logistic regression analysis of the previous section, factor effects were analyzed separately for “Day” and “Day & Night” matches now using quantile regression. Here, our data set consists of 507 ODI matches played from January 2008 to August 2011. In addition to the ICC teams, some none-ICC teams were also included for the matches they played under ODI status.

Table 6: Quantile Regression Estimates for “Day” Matches.

	$\tau = 0.15$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.85$	Mean
INTERCEPT	−102.72	−68.62	−15.71	33.40	63.00	−17.62
TOSS	28.87	10.12	6.44	11.10	14.21	14.48
HOME	37.91	25.12	30.76	16.60	11.48	25.72
SUPERIORITY	0.94	0.95	1.34	1.15	1.26	1.16

Table 7: Quantile Regression Estimates for “Day & Night” Matches.

	$\tau = 0.15$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.85$	Mean
INTERCEPT	−63.57	−46.15	−8.84	52.24	74.37	0.08
TOSS	8.24	15.75	23.24	0.94	8.59	16.68
HOME	−7.38	−8.78	−21.26	−13.24	19.13	−8.88
SUPERIORITY	1.36	1.44	1.76	1.78	1.27	1.51

The SAS[®] 9.2 quantile regression procedure was used in this analysis. Tables 6 and 7 give the parameter effect estimates for quantile regression at selected quantiles $\tau = 0.15, 0.25, 0.50, 0.75$

and 0.85 for the two ODI formats. The last columns, **Mean**, correspond to OLS regression estimates for the corresponding parameters given in the first columns of these tables. Additionally, Figures 1 – 8 provide graphical presentations of the effects of the factors for “Day” and “Day & Night” matches separately, with corresponding confidence bands.

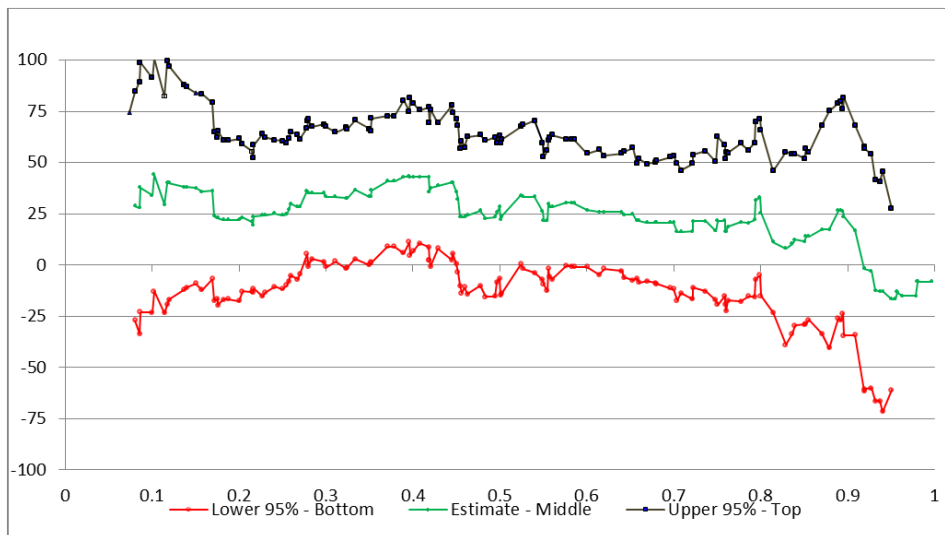


Figure 1: Home-Field Effect for “Day” Matches.

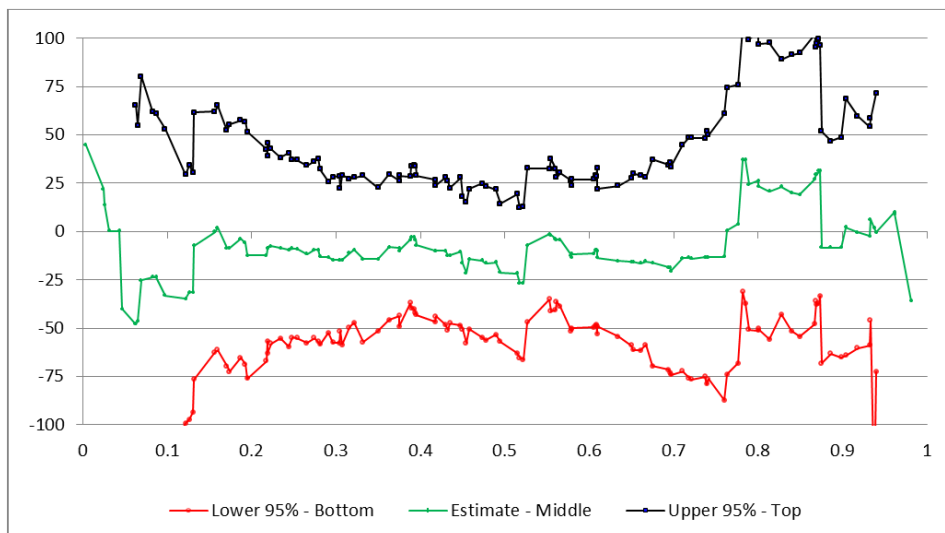


Figure 2: Home-Field Effect for “Day & Night” Matches.

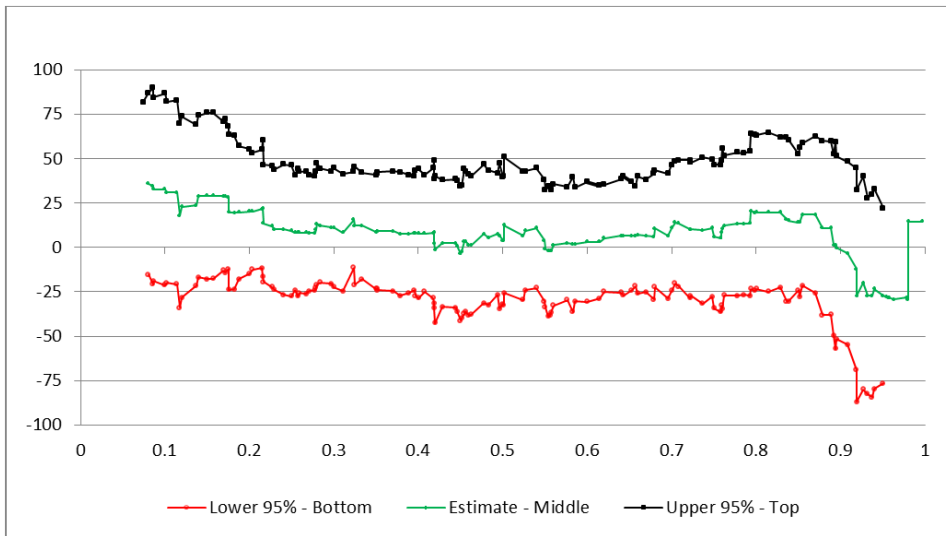


Figure 3: Toss Effect for “Day” Matches.

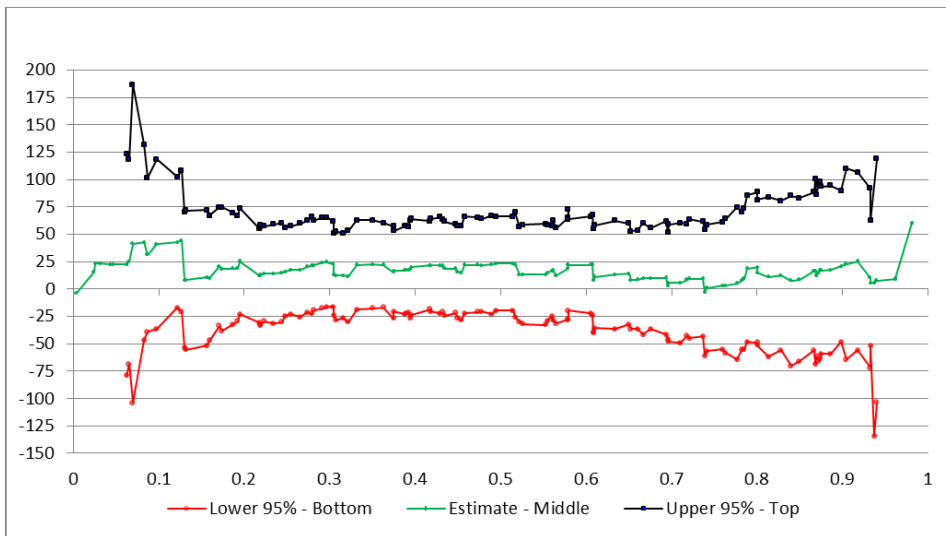


Figure 4: Toss Effect for “Day & Night” Matches.

Figures 1 and 2 show that the effect of “home-field-advantage” is clearly positive only for “Day” matches. This is consistent across most quantiles. “Day & Night” matches have become popular among cricket fans mainly due to convenience. In the past most cricket games were “Day” only, but now most of the cricket-playing countries have facilities to play “Day & Night” matches. This analysis also shows that the earlier belief, that playing at home gives an advantage, is correct, but only for

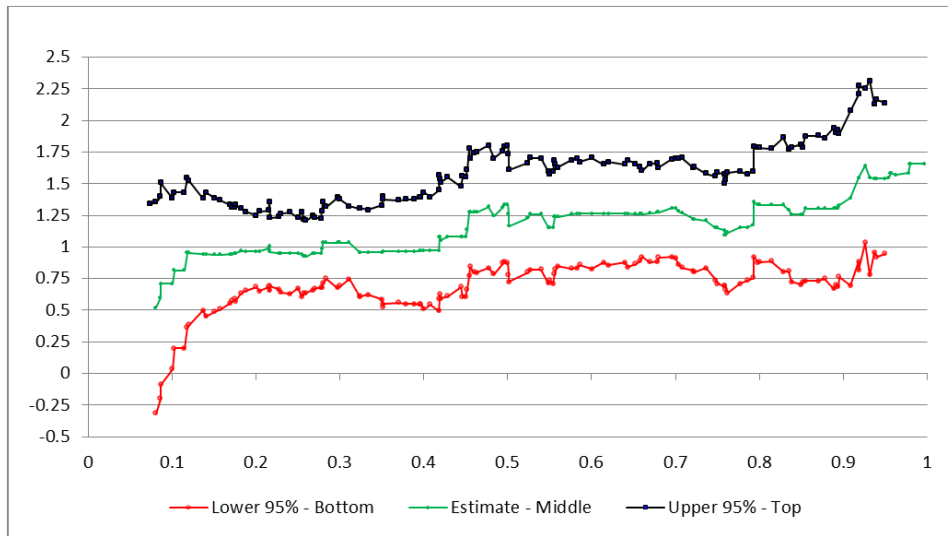


Figure 5: Superiority Effect for “Day” Matches.

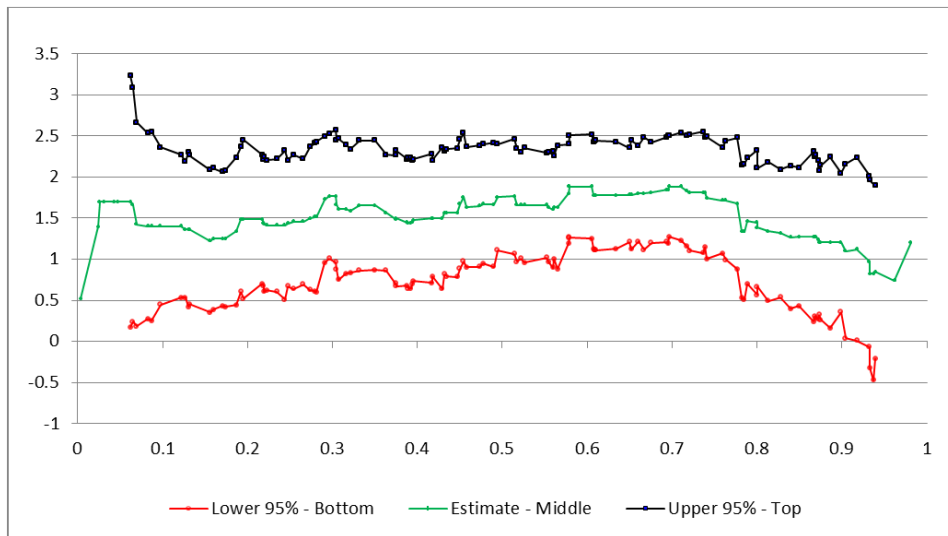


Figure 6: Superiority Effect for “Day & Night” Matches.

“Day” matches. Indeed, the final columns of Tables 6 and 7 report the OLS regression estimates for the “home-field-advantage” effect. Notice that for “Day” matches, on average, the home team scores 25.72 (P -value = 0.0548) more runs than the opposing team. However, for “Day & Night” matches the home team scores, on average, 8.88 runs fewer than the opposing team, even though this result is not statistically significant (P -value = 0.5936). Figures 3 and 4 show that the quantile regression

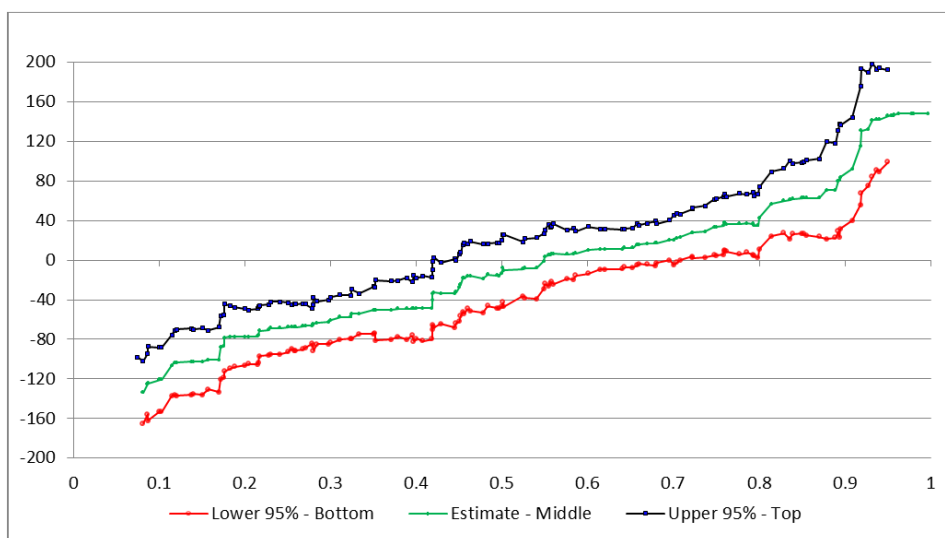


Figure 7: Intercept Effect for “Day” Matches.

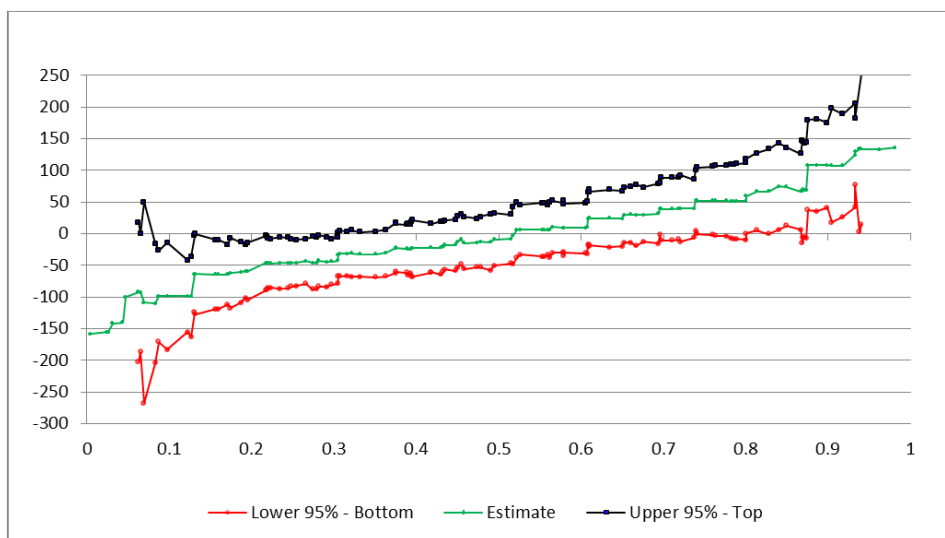


Figure 8: Intercept Effect for “Day & Night” Matches.

results for “winning the toss” are generally positive for each format of fifty-over matches, although this effect is not statistically significant. For “Day & Night” matches, environmental conditions could change drastically within a short period of time, and this could add additional influences of other factors, such as humidity, temperature, visibility, etc. that are not considered in this statistical model, but could play a greater role in deciding the match winner. Quantifying these additional fac-

tors, which could dilute the significance of the “home-field-advantage”, into the model is an arduous task. The parameter estimates in Tables 6 and 7 along with Figure 1 – 8 demonstrate the effectiveness of quantile regression in comprehensively understanding the joint effects of the factors, instead of focusing solely on the behavior of the average.

4. Discussion and Conclusions

In this paper we have investigated the effect of the “home-field-advantage” on the outcome of One Day International (ODI) cricket matches based on two different regression approaches: logistic regression and quantile regression, which is a novel approach. In addition, we have studied whether there is a significant advantage in “winning the toss” on the outcome of ODI matches. Our results have shown that the “home-field-advantage” is a significant factor only for “Day” matches. Previous studies have shown that the “home-field-advantage” is a significant factor for ODIs in general, and we have extended this result further and shown that this is, in fact, significant for “Day” matches only.

Confirming previous studies, we have also shown that “winning the toss” does not give any statistically significant advantage towards the outcome of the game. “Winning the toss” may not, in itself, be sufficient unless the team captain makes the initial right call to either “bat” or “field” first, depending on her/his judgment and playing conditions. Further analysis, which takes into account the team captains’ decisions based on the outcome of the toss, would be an excellent future study. Additionally, the “home-field-advantage” may well be dependent upon a specific opposing team. For example, India’s “home-field-advantage” against Sri Lanka when playing in India may be vastly different than when India plays against Australia in India. Sri Lankan players’ familiarity with the environmental conditions and pitches in India could dilute India’s home-field-advantage to some extent.

Teams generally decide to bat first when they win the toss mainly if they have stronger batting ability or because of the unpredictability of the second half of the game possibly being influenced by environmental conditions. In fact, this second-half unpredictability is more pronounced for “Day & Night” matches. Therefore, stronger teams generally prefer to bat first in “Day & Night” matches. The current ICC ranking system for ODI teams does not take into account whether a particular match is played at home or away, nor does it distinguish between “Day” and “Day & Night” matches. Given the fact that the “home-field-advantage” is significant only for “Day” matches, perhaps there is now justification for incorporating it into calculations of team rankings.

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